Reasoning on Quantitative Properties of Programs with an Algebraic Approach

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Education: Maths and CS

MSc in Mathematics & Computer Science *Gustave Eiffel Univ., in progress*

- Labex Bézout Scholar, ranked 1st (excluding internship)
- Relevant coursework: complexity theory, combinatorics, and discrete optimization.

MSc in Applied Mathematics *Univ. of Limoges, 2025*

- Ranked 1st
- Relevant coursework: stochastic optimization and game theory.

BEng in Computer Science (Honors Program) *Univ. of Technology, Vietnam, 2024*

- Relevant coursework: mathematical modeling (including Hoare logic and model theory for first-order logic), functional programming, and compilers.

Research Experience: Algorithms & Formal Logic

Compatibility between Open Automata and Session Types *M2 internship, Telecom Paris, 2025*

- **Core Result:** Proved that a simplified OA model is equivalent to multiparty session types, mapping safe composition to subtyping.
- **Takeaway for PhD:** Foundational experience in formal methods, including linear logic (useful for reasoning on resource usage), and establishing structural equivalences between distinct computational models.

Complexity Analysis of the Regex Matcher Hyperscan *M2 Internship, LIGM, in progress*

- **Core Analyses:** FDR string matcher (SIMD extension of Shift-Or), correctness of decomposition-based matching, and longest literal in average.
- **Takeaway for PhD:** Skills in complexity analysis, which is useful for checking examples and counterexamples towards developing a theory.

PhD Project

$$\text{Reasoning on } \underbrace{\text{Quantitative Properties}}_{\substack{\text{bounds of success/failure prob.,} \\ \text{execution time,} \\ \text{size bound of variables}}} \text{ of Programs with an } \underbrace{\text{Algebraic Approach}}_{\text{Kleene algebra with tests}}$$

A Kleene algebra with tests (KAT) is a Kleene Algebra (an axiomatization of regular expressions) extended with a Boolean subalgebra.

Programs can be encoded as KAT expressions:

$$\begin{aligned} \textbf{if } b \textbf{ then } p \textbf{ else } q \quad &\text{as} \quad bp + \overline{b}q \\ \textbf{while } b \textbf{ do } p \quad &\text{as} \quad (bp)^* \overline{b} \end{aligned}$$

For example we can prove Loop Unrolling equivalence

$$\textbf{while } b \textbf{ do } p \quad \equiv \quad \textbf{if } b \textbf{ then } (p; \textbf{while } b \textbf{ do } p) \textbf{ else skip}$$

by proving algebraically

$$(bp)^* \overline{b} = bp((bp)^* \overline{b}) + \overline{b}. \text{ (factor RHS, then use } 1 + xx^* = x^*)$$

PhD Project - Contexts and Motivations

KAT subsumes the traditional *Hoare logic* (HL) in checking input-output properties (*what* a program does) [1].

KAT is worth studying for its convenience in checking equivalence (shown above). For HL, given C_1 and C_2 one has to prove $\{P\}C_1\{Q\}$ and $\{P\}C_2\{Q\}$ separately for every properties P and Q.

HL has been extended to *Graded* HL for intensional, quantitative properties (*how*) e.g., $\{P\}C\{Q\}_\beta$ (starting with P satisfied, executing C leading to $\neg Q$ satisfied with probability at most β) [2].

KAT has been optimized and extended successfully to approximate Guarded KAT (aGKAT) for probabilistic properties [3].

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[1] Kozen, D. (2000). On Hoare logic and Kleene algebra with tests.
[2] Gaboardi, M. et al. (2021). Graded Hoare logic and its categorical semantics.
[3] Gomes, L., Baillot, P., & Gaboardi, M. (2025). A Kleene algebra with tests for union bound reasoning about probabilistic programs.

PhD Project - Goal and Fruitfulness

Goal: Drawing from KAT and Graded HL, to develop a general framework of *Graded KAT* to formalize quantitative resource bounds.

Fruitfulness:

- Theoretical advancement.
- Targeting practical coding patterns (programs are in polynomial time, `for` loops are preferred when possible), a matured theory can be integrated into automated verification tools (e.g., a linter).

PhD Project - Challenges

What is common among the desired properties? Relations are not always trivial, e.g.,

- In analysis of algorithms, the number of comparisons of a sorting algorithm intuitively relates to its execution time.
- It is not the case for lower bound of success/failure probability and upper bounds [4].

With a group of properties that have a common structure, how to design axioms and a satisfying semantics?

This is an iterative progress.

[4] Gomes, L., Baillot, P., & Gaboardi, M. (2025). A Kleene algebra with tests for union bound reasoning about probabilistic programs.

PhD Project - Plan

First phase (6 months): Drawing from GKAT, to develop a theory for execution time, by proposing axioms and experimenting with examples and counterexamples.

Second phase (9 months): To develop a theory that subsumes the first phase and aGKAT.

Third phase (6 months): To develop a semantics for the theory in the second phase, drawing from the categorical semantics of Graded HL.

Long-term: To iterate when other resource bounds are considered. Then to adapt for relational properties (e.g., equivalence).

Thank you for attention !